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Vision-Based Lane Inference for Unstructured and Structured Indian Roads

BSc Computer Science · 2025-26

Under the guidance of

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Problem Statement

Lane detection assumes paint. Most of the Indian network has none.

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What every method assumes

THE SHARED PREMISE

- Painted lane markings are present in the frame
- Trained and evaluated on TuSimple, CULane, BDD100K
- All of them marked highways and marked cities

What Indian roads give you

THE ACTUAL INPUT

- Paint is absent, worn, or hidden under traffic
- Yet the road still carries a structure drivers agree on
- The information is in the geometry, not the markings

And the failure is silent

A detector that finds nothing looks exactly like one that failed. Nothing downstream can tell the two apart, so the roads where lane information would help most are the ones that get none.

That is why this project infers structure instead of detecting it.

Objectives & Scope

Rebuild the pipeline so every claim it makes can be measured.

Objectives

WHAT TO BUILD

- Train a segmentation network on Indian road imagery
- Recover metric geometry with pitch measured, not assumed
- Infer a lane count from measured width and a design standard
- Report a confidence that predicts error
- Measure against baselines, including trivial ones

Scope

BOUNDARIES

- In: monocular forward-facing camera
- In: single image, video clip, live camera
- In: lane count and metric widths, both validated
- Also reported, not validated: ego lane, road-user distances
- Out: path planning and vehicle control
- Out: any sensor other than the camera
- Out: night and heavy rain

METHODS THIS RESTS ON

MobileNetV3 | LR-ASPP | FPN | IPM | Kalman filter | IRC:86 | Knowledge distillation

Existing Work

Mature field, built on an assumption this project cannot make.

Segmentation

SCNN · LANENET

- Label marking pixels, then cluster them into lane instances

Row anchor

ULTRA FAST LDD

- Select which cell in each image row holds a marking

Parametric

POLYLANENET · CLRNET

- Regress lane curves directly from features

The limitation they share

Each assumes markings exist. On an unmarked road the output is empty or arbitrary, and nothing in the formulation distinguishes the two.

Why IDD cannot fix it

NO LANE GROUND TRUTH

- IDD annotates drivable surface, not lanes
- An unmarked road has no lane to annotate
- So structure has to be inferred, not learned

System Design

Four tiers, one API surface, and one piece of state that matters.

Client

BROWSER AND CAMERA

- React interface, mobile camera, file upload

Server

FASTAPI

- Eleven endpoints. Every request enters the same way

Inference

STATELESS MODULES

- Segmentation, geometry, temporal tracker

Storage

READ ONLY

- Weights, sample data, measured results

The one piece of state, and what it costs

Temporal smoothing needs the previous frame, so filter state is kept per client between requests. That is why the system needs a long-running process and cannot be deployed as stateless serverless functions.

STACK USED

Python 3.12 · PyTorch 2.13 · OpenCV 5.0 · FastAPI · Uvicorn · React 19 · Vite 8 · Tailwind · pytest

Tools & Technologies

Everything below is pinned in the repository and reproducible.

Model

3.33 M PARAMETERS

- MobileNetV3-Large backbone
- LR-ASPP context head, FPN decoder
- Seven semantic classes plus a lane head

Pipeline

GEOMETRY AND TRACKING

- Inverse perspective mapping
- Occlusion-bridged width profile
- Kalman filter and majority vote

Delivery

A RUNNING APPLICATION

- FastAPI backend, eleven endpoints
- React interface, report and demo routes
- 32 tests over geometry, metrics and API

STACK USED

Python 3.12 | PyTorch 2.13 | torchvision 0.28 | OpenCV 5.0 | NumPy 2.5 | FastAPI 0.141

Uvicorn | React 19 | Vite 8 | Tailwind 3 | Recharts | pytest | Apple MPS

127 min
training, 40 epochs on the laptop GPU

36.5 ms
end to end per frame, 27 fps

32
tests, all passing

Implementation

Four stages, and the defect each one exposed when it was measured.

Segmentation

ONE FORWARD PASS

- Two heads share the decoder, so lane markings cost no extra time

Geometry

PITCH PER FRAME

- Clipped edge points excluded. One frame gave 25.1° of pitch until they were

Measurement

SEE THROUGH TRAFFIC

- Occlusion handled per row. Bridging every row doubled the road across a median

Refusal

DECLINE, DO NOT GUESS

- The scene is checked first. The network calls a black frame 26% drivable

The largest single defect we found

Bridging vehicles into the road mask fixes parked vans and breaks medians. Choosing per row instead took relative width error at p90 from 151% to 47%. We found it by checking the lane model against the mask it was built from, not against ground truth.

SUPPORTING PIECES

- CIELAB fallback
- hole filling
- bottom-connected component
- polynomial edge fit
- per-state Kalman noise

Results & Analysis

Measured on the 204-frame IDD-Lite validation split.

0.9310
drivable IoU, deployed

0.7754
boundary F1, deployed

165/204
lane solutions, 80.9 per cent

27 fps
end to end, 36.5 ms per frame

Against what it replaces

THE FLOOR

- Study project pipeline 0.4270 drivable IoU
- Fixed trapezoid, which ignores the image, 0.6450

Ceiling versus deployed

AN HONEST DISTINCTION

- The segmentation head alone reaches 0.9403
- Mask cleanup costs 0.0093 IoU and buys three more lane solutions

Geometry accuracy

AGAINST GROUND TRUTH EDGES

- Median road-edge error 0.31 m
- Over 2,288 samples at 6 to 22 m ahead

The result that reframed the project

The pipeline this replaces scored below a fixed trapezoid that never looks at the image. That is what showed the problem was structural, not a tuning problem.

Challenges & Limitations

Stated here rather than left for the viva to find.

One metric assumption

CAMERA HEIGHT 1.75 M

- Every distance scales linearly with it
- One known width removes it entirely

Unvalidatable interior

NO GROUND TRUTH EXISTS

- Carriageway extent is validated
- The divisions inside it cannot be

Half the data unused

A PREPARATION BUG

- Part II of IDD-20k was downloaded and never merged
- 6,993 of about 14,000 frames

Low light, wide angle

OUT OF DISTRIBUTION

- Trained on daylight, normal-lens footage

One run, one seed

NO INTERVALS

- Small differences between configurations are not significant

Optimistic epoch choice

VALIDATION ONLY

- Best epoch picked on the split it is reported on, since the test split is unlabelled

ALSO OUTSTANDING

- ego lane unvalidated
- road-user distance unvalidated
- wrong-side detection off by default
- narrow-road divisor to confirm

Conclusion & Future Work

The idea holds, and it is now measurable rather than plausible.

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Conclusion

WHAT WAS SHOWN

- Lane structure can be inferred from road geometry where no paint exists
- 0.9310 drivable IoU deployed, 165 of 204 frames solved, 0.31 m median edge error, 27 fps
- The harness and its trivial baselines are what made every other result interpretable

Future work

IN ORDER OF VALUE

- Merge Part II of IDD-20k and retrain
- Train on IDD-AW for low light and adverse weather
- Retrain at 384 by 640 for thin markings
- Validate ego lane and road-user distances
- Confirm the narrow-road divisor against the standard

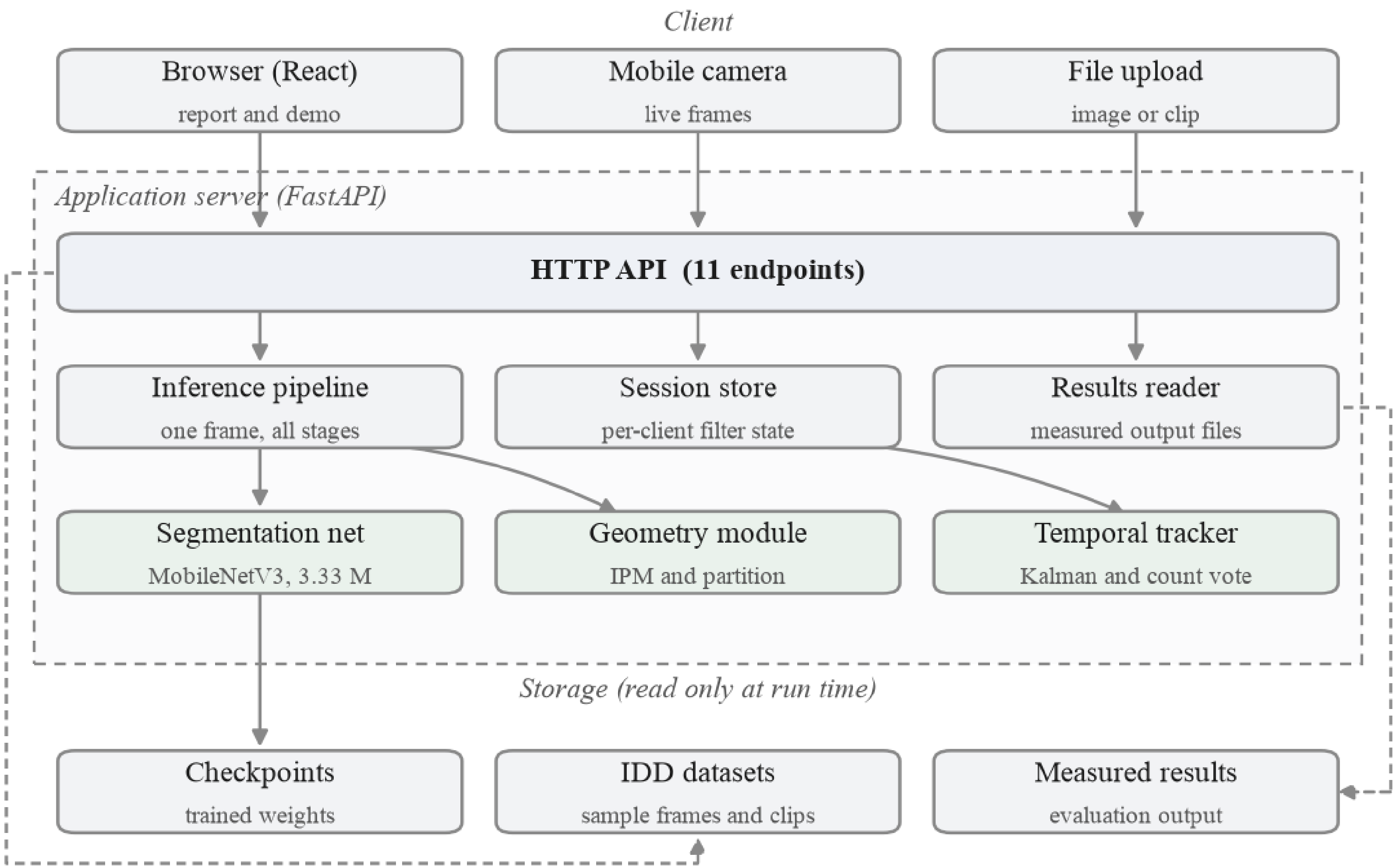
4
hypotheses tested and rejected

3
earlier claims reversed

20
figures, all from source

System Architecture

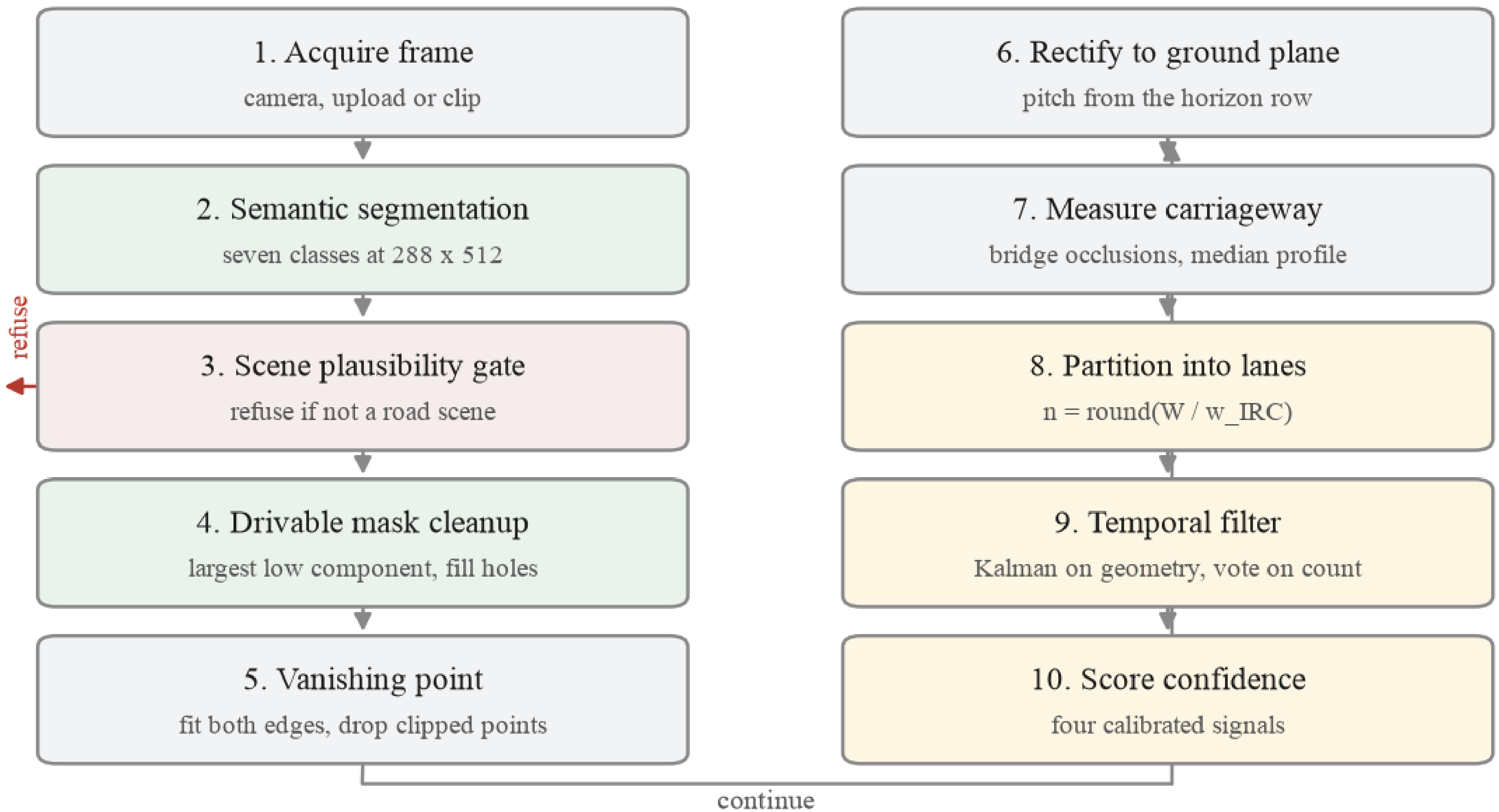
Four tiers. Everything inside the dashed box is one process.



End-to-End Process Flow

Ten stages, including the path where the system declines to answer.

Every stage writes an intermediate image, which is what the stage walkthrough in the demo displays.



Metric Scale, No Calibration

The dataset ships no camera metadata. It turns out not to need any.

$$dX = du \cdot H / (v - v_{horizon})$$

Lateral ground scale from image row alone. The focal length cancels.

Pitch is measured

NOT ASSUMED

- Recovered per frame from the vanishing point row

Why it cancels

THE HORIZON CARRIES IT

- The horizon row already encodes $f \cdot \tan(\theta)$, the same product that sets ground scale

What is left

ONE ASSUMPTION

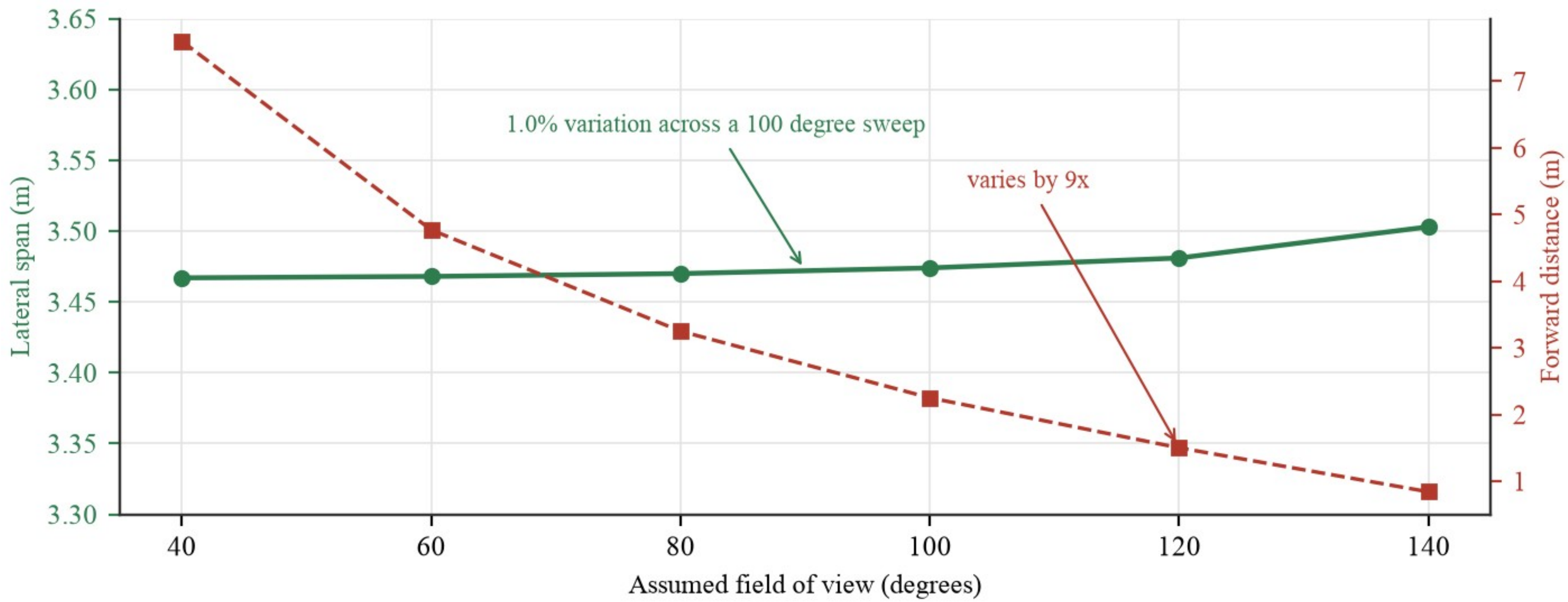
- Lateral scale moves 1.0% across a 40° to 140° sweep. Only camera height remains

HOW IT IS RECOVERED

Inverse perspective mapping | Vanishing point | Pitch ±15° | First-order error < 1.6%

Focal Length Cancels

Rebuilding the transform across a hundred degrees of assumed field of view.



From Width to Lane Count

A published design width applied to a measurement, not an arbitrary split.

n = round(W / w)

W is the measured carriageway in metres, w the design lane width.

Choosing w

A THRESHOLD ON WIDTH

- 3.50 m when the carriageway is 6.0 m or wider
- 3.00 m below that
- A rule about width, not a judgement about the road

A floor at 2.5 m

AND A REFUSAL

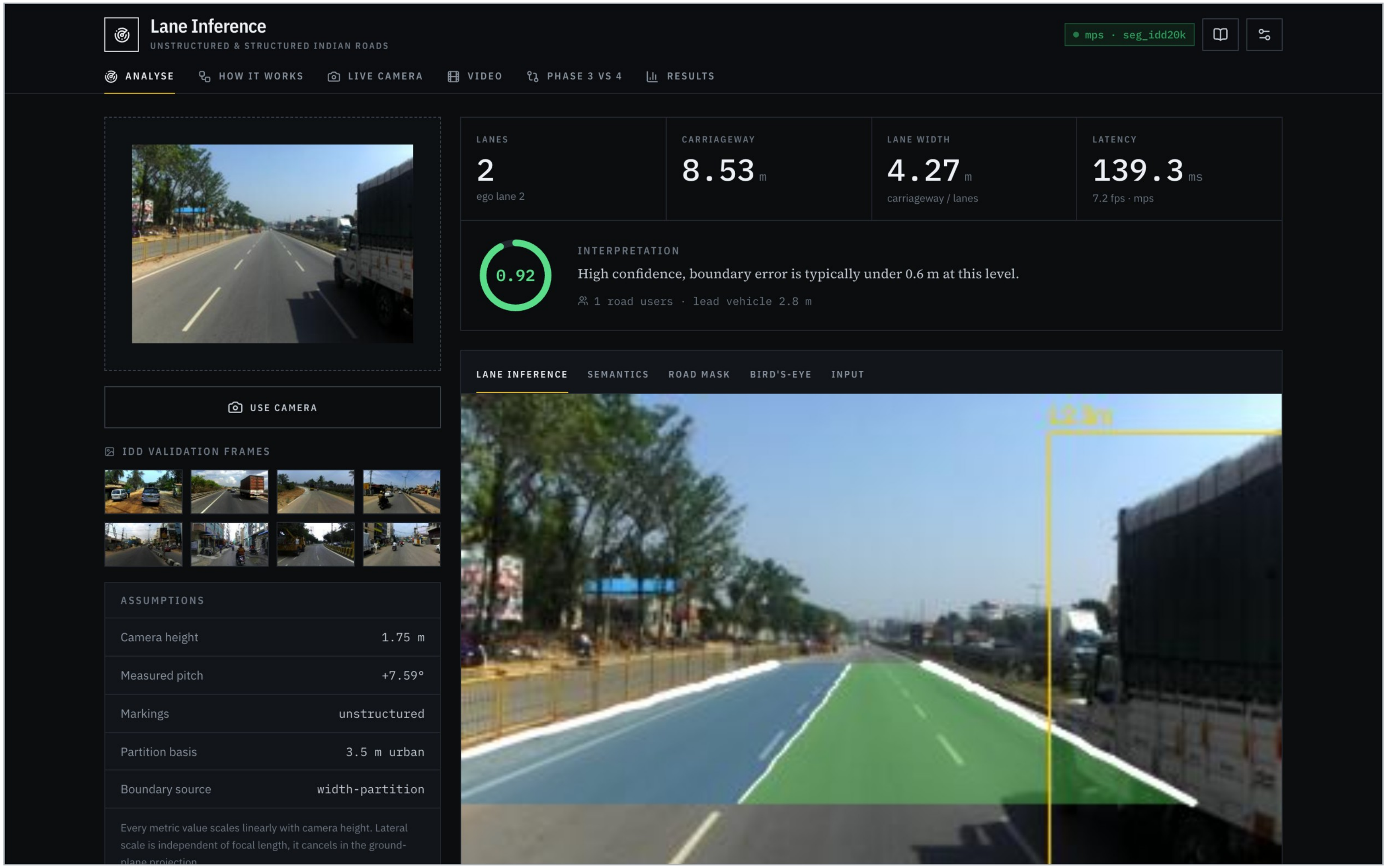
- If the division gives narrower lanes, the count drops
- If even one lane falls below the floor, the frame is refused
- That is the largest single refusal cause

THE PARTITION RULE

- IRC:86 urban 3.50 m
- narrow road 3.00 m
- floor 2.50 m
- occlusion bridging
- median profile

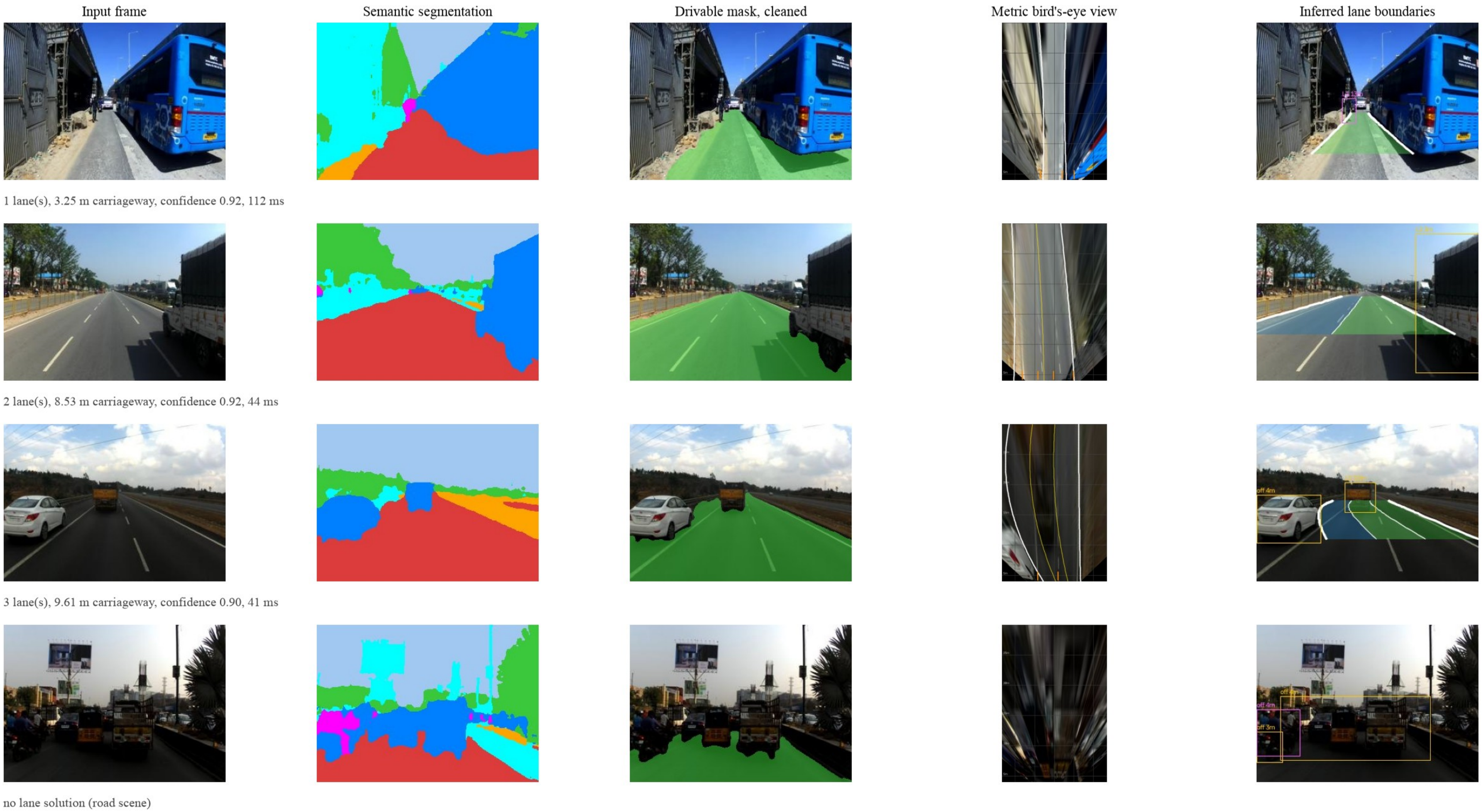
The Application

Two routes: the report, and a demo running the real pipeline.



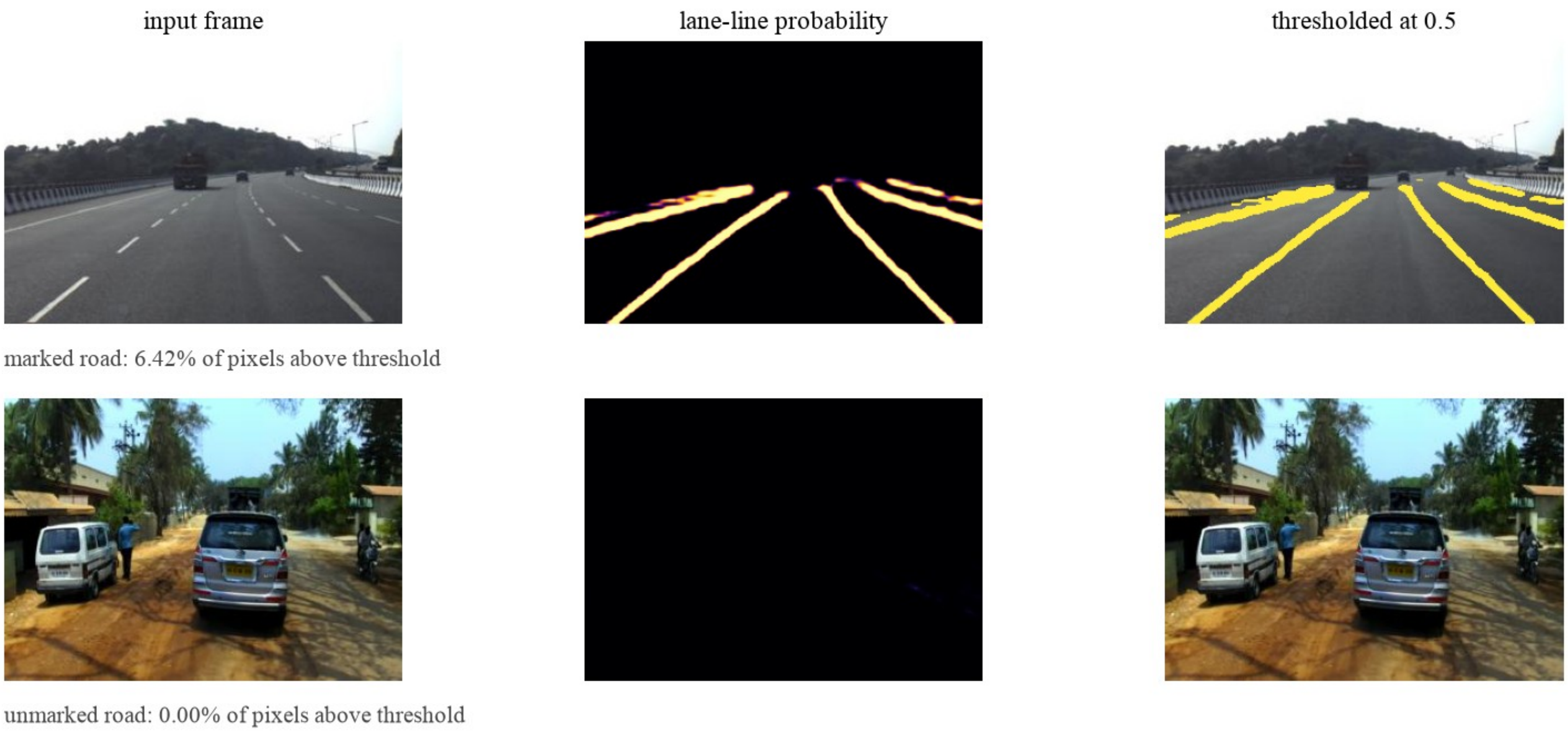
Pipeline Stages

A narrow road, a two-lane road, a painted three-lane road, and a refusal.



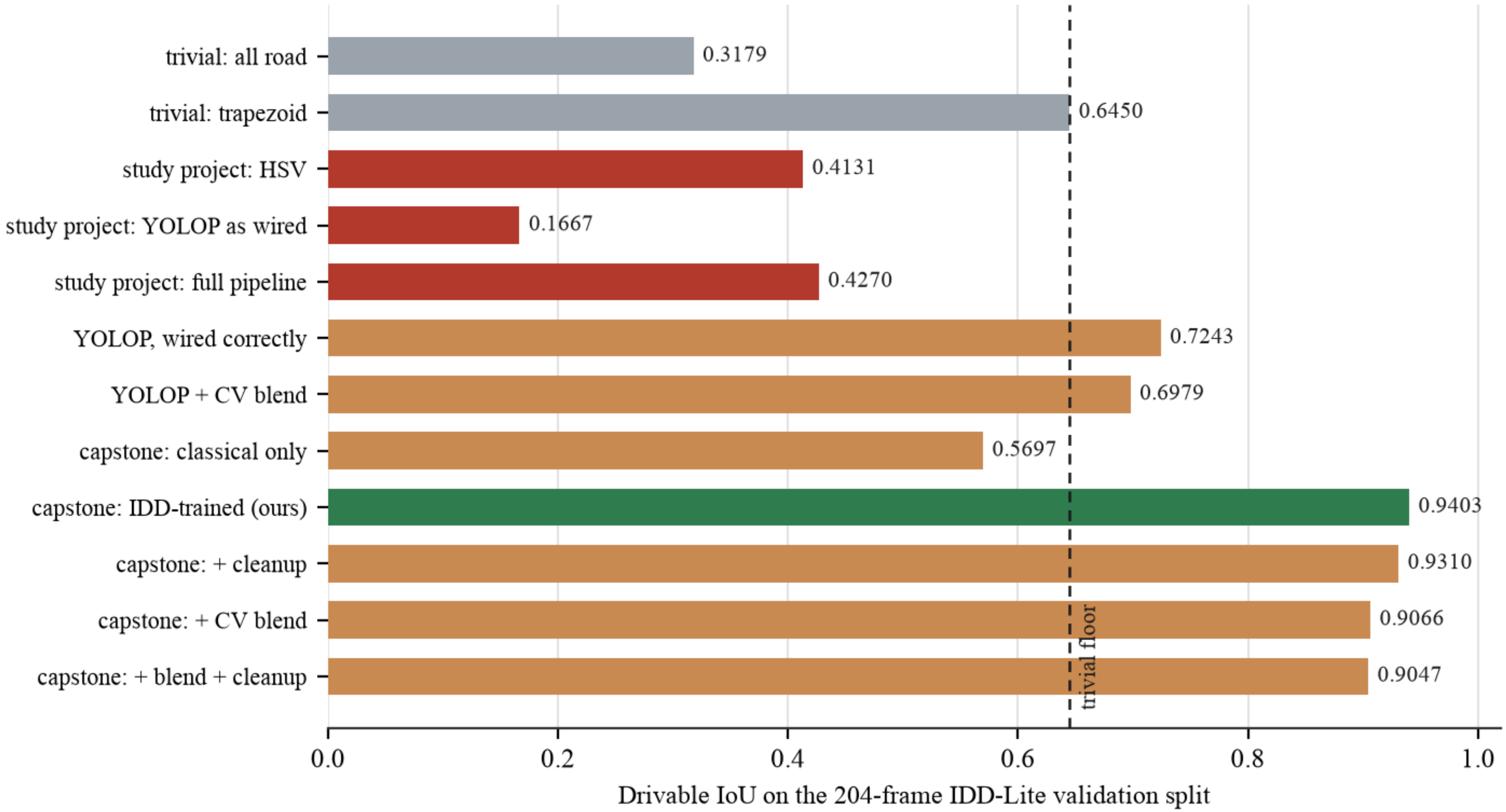
The Lane-Marking Head

IDD has no lane annotation, so YOLOP was used offline as a teacher.



Measured Against a Floor

Twelve configurations, same frames, same harness. Two of them never look at the image.



Is the Geometry Correct?

The hardest question in the project, and the easiest one to avoid.

There is no lane truth

THE CONSTRAINT

- No Indian dataset annotates lanes
- That absence is the premise of the project

But edges can be checked

WHAT IS VALIDATABLE

- The outermost inferred boundaries must match the real road edges
- Drivable ground truth gives those

And the tail is heavy

STATED PLAINLY

- Median 0.31 m, but p90 approaches 4 m
- Relative width error p90 is 151 per cent

So the confidence score was rebuilt

Correlation with measured error 0.272, against 0.187 before. Gating at 0.7 keeps 68 per cent of frames and removes 21 per cent of the median error.

Still not validated

INHERENT TO THE PROBLEM

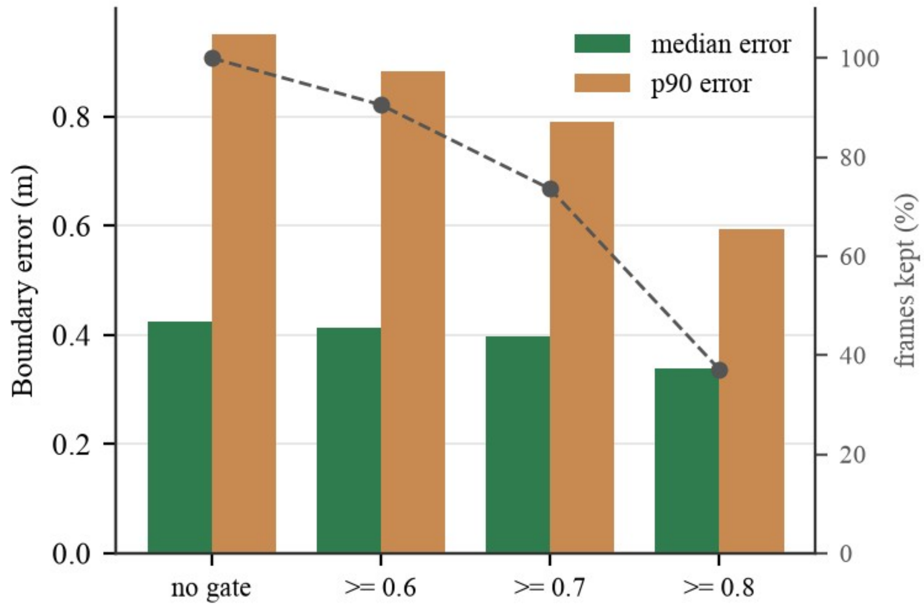
- Interior lane divisions on unmarked roads
- No ground truth exists anywhere for them

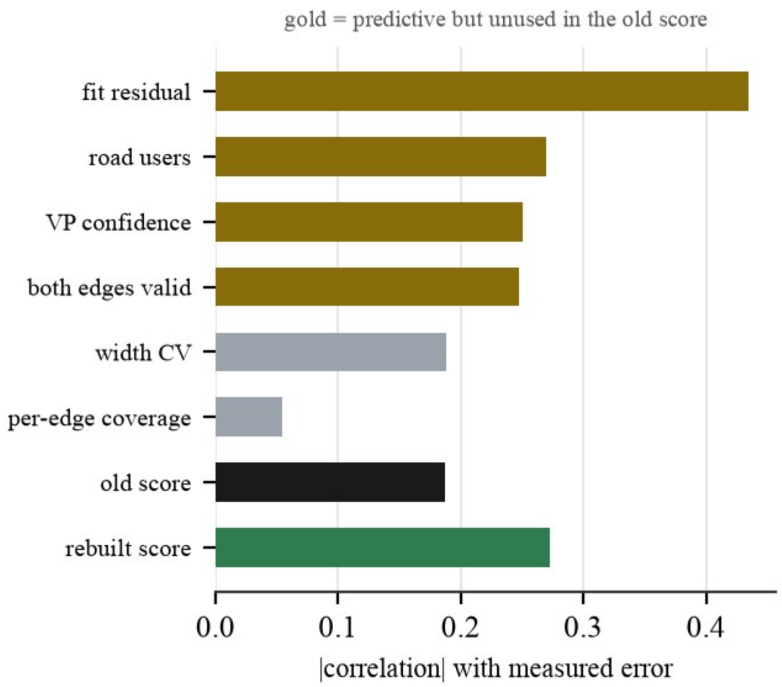
SIGNALS THE REBUILT SCORE USES

fit residual +0.435 | **VP confidence −0.251** | **both edges valid −0.248** | **width CV +0.188**

Confidence That Predicts Error

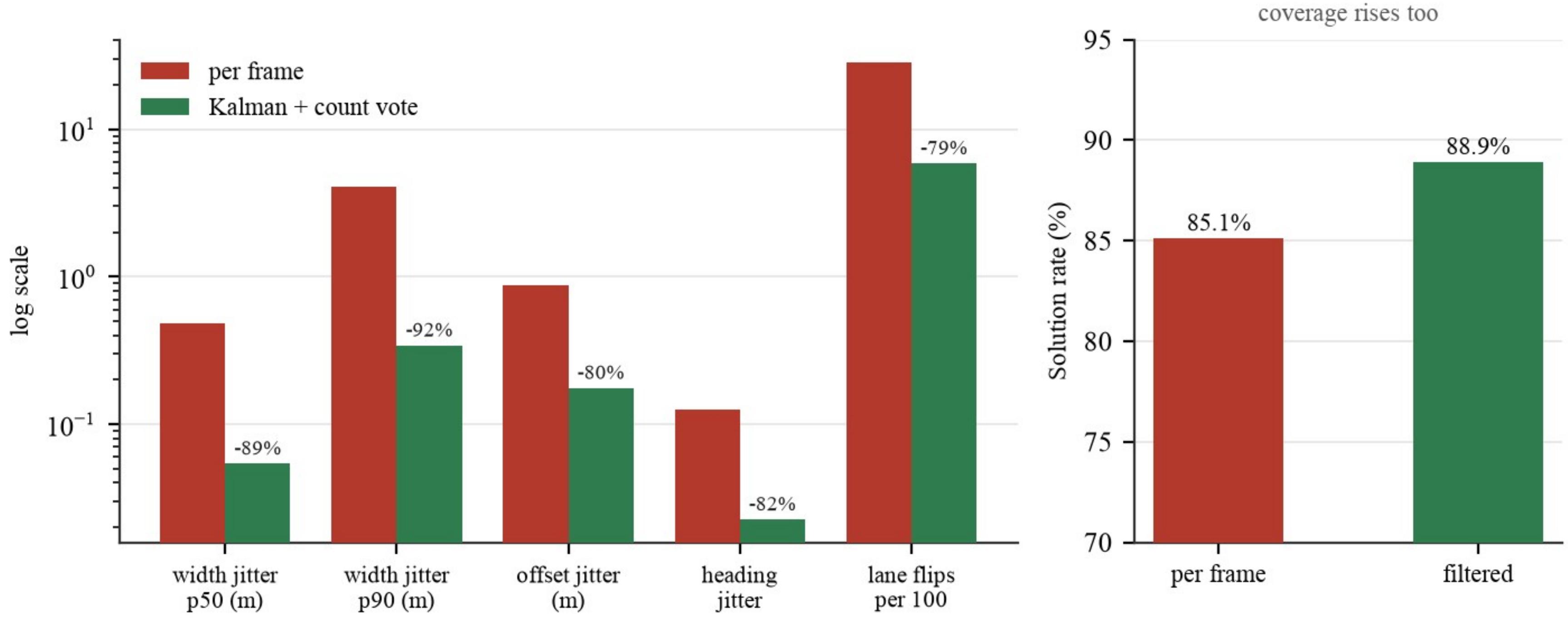
Gating on the rebuilt score removes most of the error tail.





Temporal Stability

150 sequences, 4,464 frames, drawn from three drives.



What Did Not Work

Four hypotheses implemented, measured and rejected.

Fusing classical and learned

0.9403 → 0.9066

- The hybrid premise the project started from is not supported by the data

Bridging before the vanishing point

90.2% → 74.5%

- Unioning vehicles recreates the frame-border clipping the fit already excludes

Lanes from observed traffic

AGREEMENT FLAT AT 34 TO 40%

- More evidence does not tighten the spacing. The modes are real and they are not lanes

Closing the dashcam gap at test time

FOUR REMEDIES, ALL REJECTED

- CLAHE works and cannot be applied: no image statistic separates the frames that need it

Measured self-correction

Three claims made confidently in earlier drafts did not survive re-measurement and are reversed in print.

Thank You

Questions

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CAPSTONE PROJECT

- Harshwardhan Mukund Mohadikar · 2023EBCS353
- Shreyas Bhat K · 2023EBCS460
- Supervised by Dr. Ashok Yemineni

Everything here is reproducible

Every figure comes from the evaluation harness in eval/ and can be regenerated from source. No number in this deck is typed in by hand.